

Secs 3.1 & 3.2- Discrete Random Variables

Intuitively a random variable X (r.v.) associates a value (a number) to every possible outcome in a Sample space S .

Definition (Random variable)

Given an experiment with sample space S ; a random variable X (r.v.) is a function from the sample space S to the real numbers $\mathbb{R}$.

Notation: random variable X numerical value x

Note: You can define multiple random variables over the same sample space

Note: A function of one or several random variables is a random variable

Eg 1. Consider an experiment where we toss a fair coin twice.

$$S = \{HH, HT, TH, TT\}.$$

Here are some random variables on this space

- (a) Let X be the number of Heads.
- (b) Let I be 1 if the first toss is tail and 0 otherwise.

$$X = 0, 1, 2$$

$$I = \begin{cases} 1 & \text{if first toss is head} \\ 0 & \text{otherwise} \end{cases}$$

Definition (Discrete and continuous random variables)

– A random variable X is called discrete if only takes discrete set of values. X can take finitely many values, or it can take infinite, but countable number of values.

– A random variable Y is called continuous if it takes continuous set of values.

Eg 2. Anything that is measured like weight, height, time are continuous random variables.

X = Wait time until a bus arrives in a bus station. **Continuous**

Y = Height of students at UMD **Continuous**

Z = Number of trials until 10 success occurs. **discrete**

H = Number of white balls in 5 draw without replacement from a urn with 8 white and 20 nonwhite balls. **discrete**

Definition (Probability mass function (pmf))

The probability mass function (pmf) of a discrete r.v. is defined for every numerical value x by

$$\begin{aligned} p_X(x) &= P(X = x) = P(\text{Event that } X \text{ equals } x) \\ &= P(\{s \in S : X(s) = x\}) \end{aligned}$$

Note: Properties of probability mass function $p_X(x)$

$$p_X(x) \geq 0$$

$$\sum_x p_X(x) = 1$$

Note that $p_X(x)$ is **the exact probability** of the event that random variable $X = x$.

pmf is only defined for discrete random variables.

Eg 2. Consider the experiment of two rolls of a tetrahedral die.

X = First roll, Y = Second roll, $Z = X + Y$.

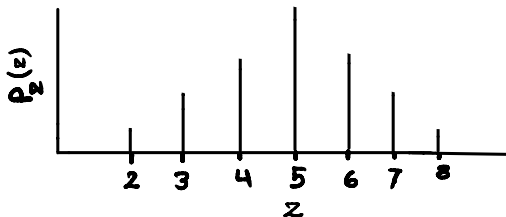
Find PMF Z .

tabular

z	2	3	4	5	6	7	8
$P(z)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

$z=2$	(1,1)	(1,2)	(1,3)	(1,4)
$z=3$	(2,1)	(2,2)	(2,3)	(2,4)
$z=4$	(3,1)	(3,2)	(3,3)	(3,4)
$z=5$	(4,1)	(4,2)	(4,3)	(4,4)
$z=6$				
$z=7$				
$z=8$				

graphical
representation



Definition (cumulative distribution function (CDF))

The cumulative distribution function (cdf) $F(x)$ of a discrete rv variable X with pmf $p(x)$ is defined on the whole real line by

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} P(y) \quad \text{for every } x \in \mathbb{R}$$

For any number x , $F(x)$ is the probability that the observed value of X will be at most x .

Eg 3. Experiment 3 toss of a biased coin with $P(H)=\frac{1}{3}$. Define the random variable

X = number of heads in this experiment.

Find PMF and CDF of X .

X can take values 0, 1, 2 and 3.

$$P_X(0) = P(X=0) = \left(\frac{2}{3}\right)^3 = \frac{8}{27}$$

$$P_X(1) = P(X=1) = \binom{3}{1} \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^2 = 3 \cdot \frac{4}{27} = \frac{12}{27}$$

$$P_X(2) = P(X=2) = \binom{3}{2} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) = 3 \cdot \frac{1}{9} \cdot \frac{2}{3} = \frac{6}{27}$$

$$P_X(3) = P(X=3) = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$$

we will finish this example on Tuesday.

$$F_X(0) = P(X \leq 0) = \frac{8}{27}$$

$$F_X(1) = P(X \leq 1) = \frac{8}{27} + \frac{12}{27} = \frac{20}{27}$$

$$F_X(2) = P(X \leq 2) = \frac{20}{27} + \frac{6}{27} = \frac{26}{27}$$

$$F_X(3) = \frac{26}{27} + \frac{1}{27} = \frac{27}{27} = 1$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{8}{27} & 0 \leq x < 1 \\ \frac{20}{27} & 1 \leq x < 2 \\ \frac{26}{27} & 2 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

Properties of CDF $F_X(x)$ for the discrete random variable X :

(1) $0 \leq F_X(x) \leq 1$

(2) $F_X(x)$ is a non-decreasing function of x

(3) $P(a < X \leq b) = F_X(b) - F_X(a)$

(4) $P(a \leq X \leq b) = F_X(b) - F_X(a^-)$.

(5) $P_X(a) = F(a^+) - F(a^-)$

In the above box

$$F(a^+) = \lim_{x \rightarrow a^+} F_X(x)$$

$$F(a^-) = \lim_{x \rightarrow a^-} F_X(x)$$

Eg 4. Assume a CDF of random variable X is given as below,

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 0.2 & 0 \leq x < 1 \\ 0.48 & 1 \leq x < 2 \\ 0.72 & 2 \leq x < 3 \\ 0.88 & 3 \leq x < 4 \\ 0.92 & 4 \leq x < 5 \\ 1 & x \geq 5 \end{cases}$$

Find

$$(a) P(1 < X \leq 3) = F(3) - F(1) = 0.88 - 0.48 = 0.4$$

$$(b) P(1 \leq X \leq 3) = F(3) - F(1^-) = 0.88 - 0.2 = 0.68$$

$$(c) P_X(1) = F(1^+) - F(1^-) = \lim_{x \rightarrow 1^+} F(x) - \lim_{x \rightarrow 1^-} F(x) = 0.48 - 0.2 = 0.28$$

$$(d) P_X(3) = F(3^+) - F(3^-) = \lim_{x \rightarrow 3^+} F(x) - \lim_{x \rightarrow 3^-} F(x) = 0.88 - 0.72 = 0.16$$